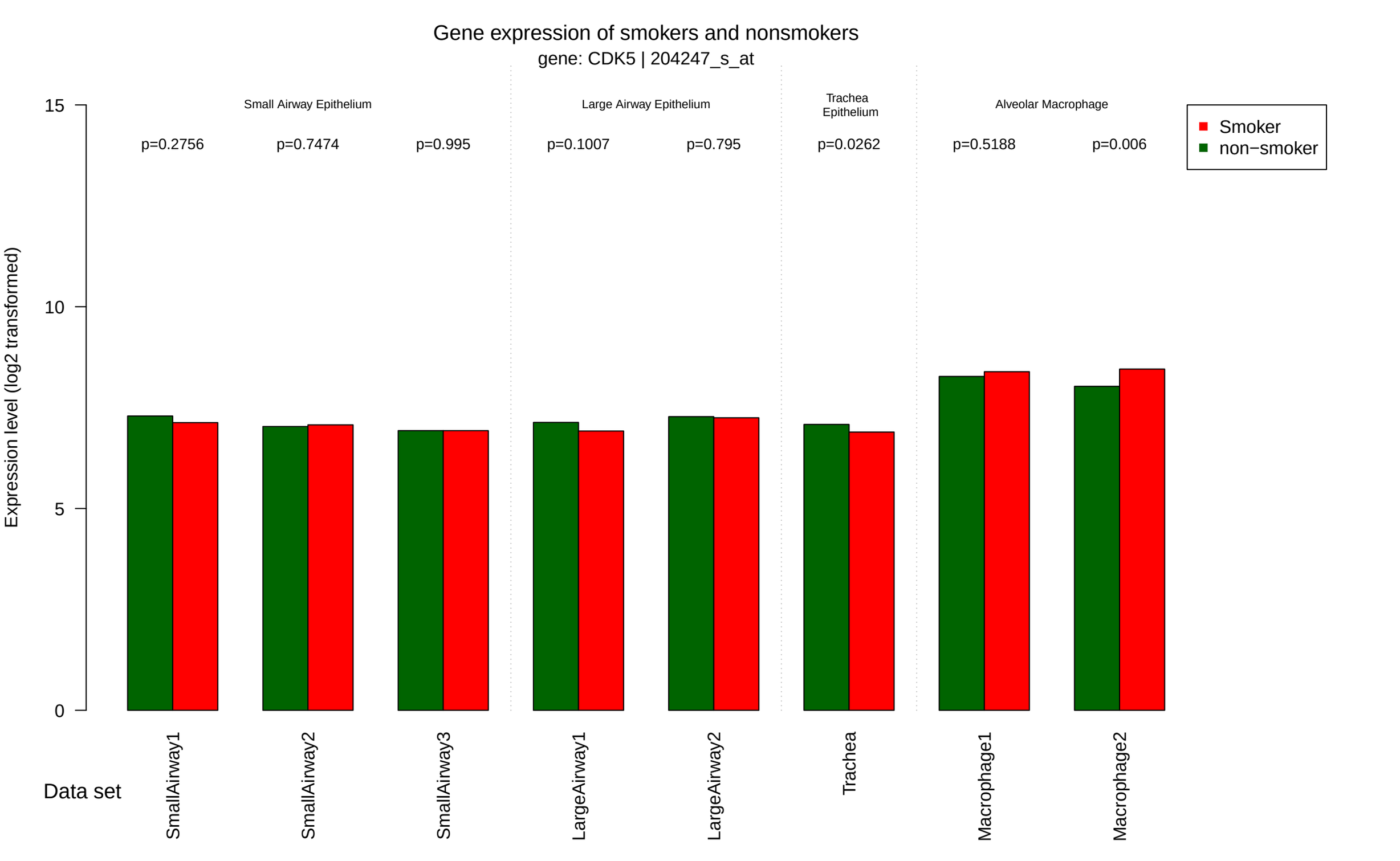
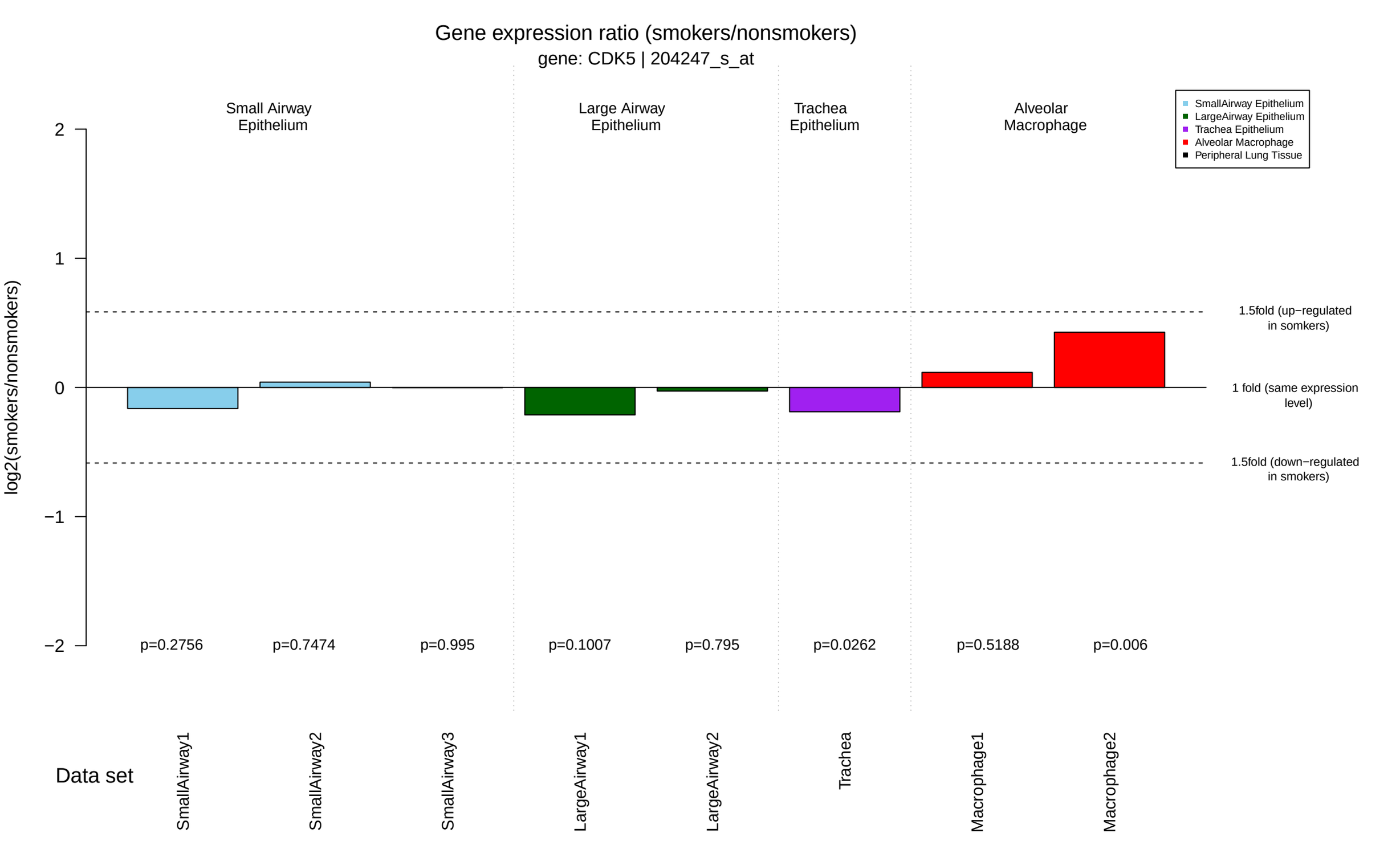
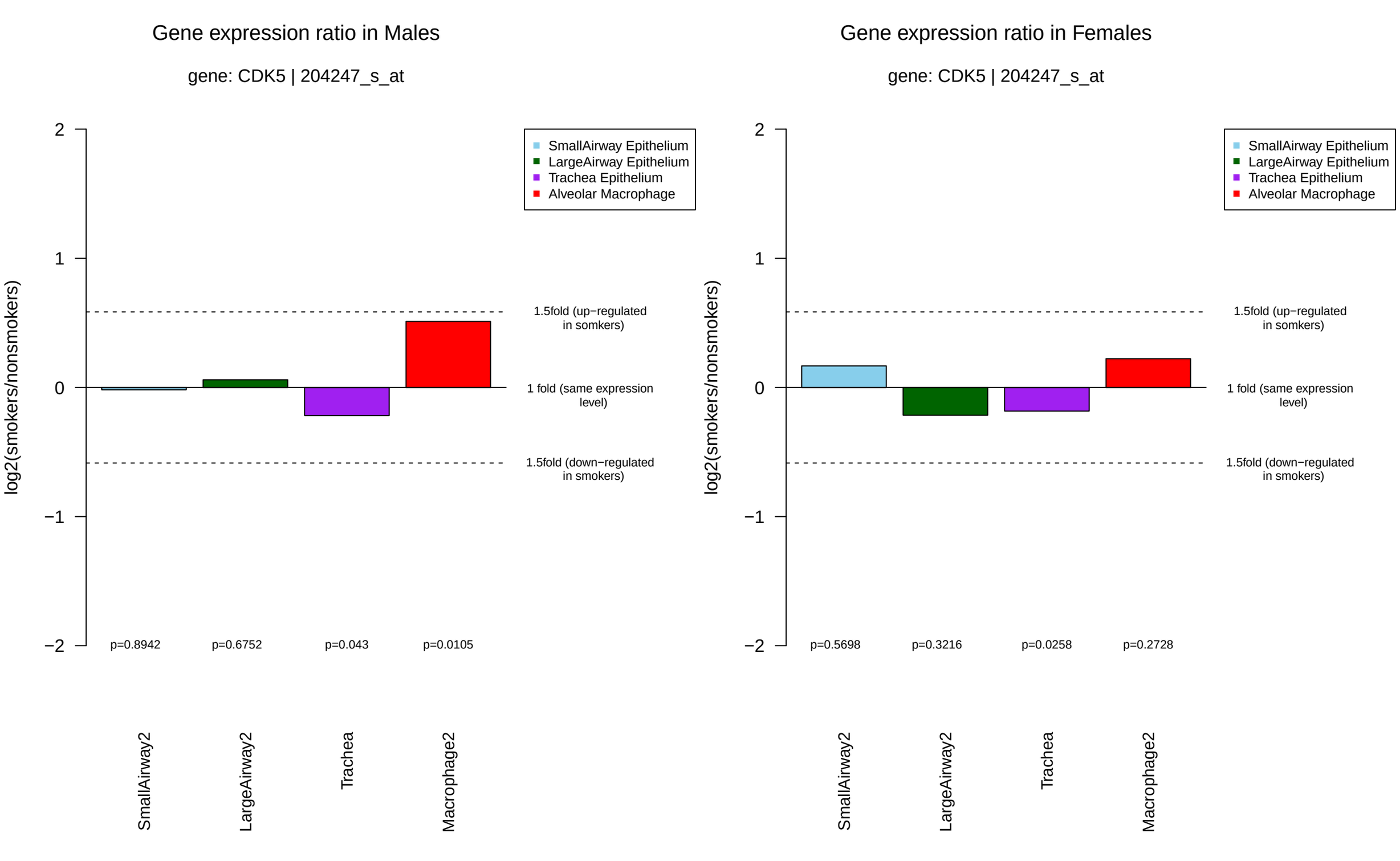
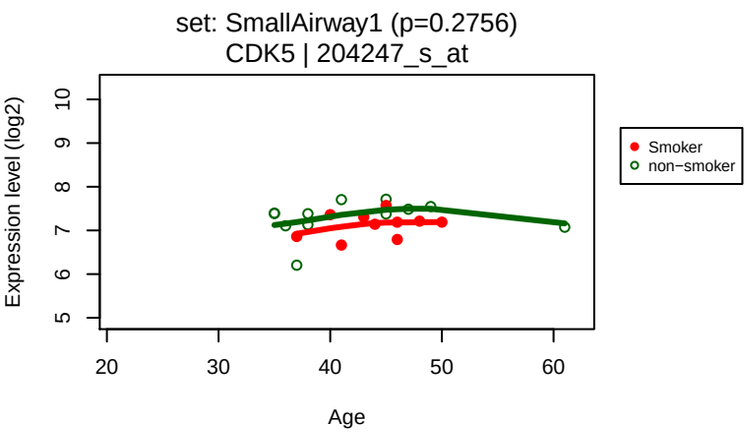
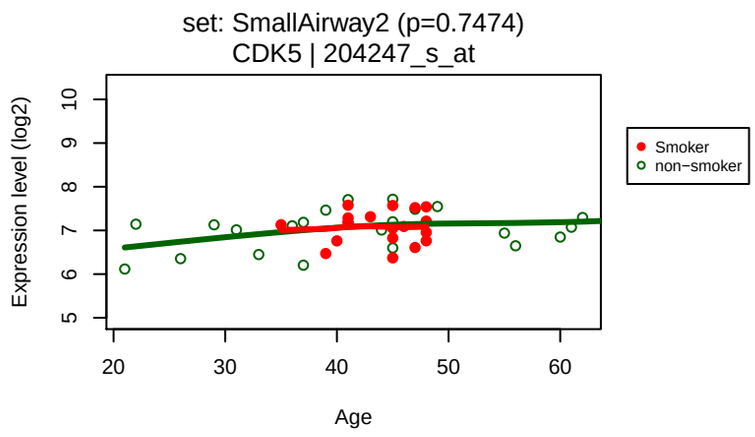
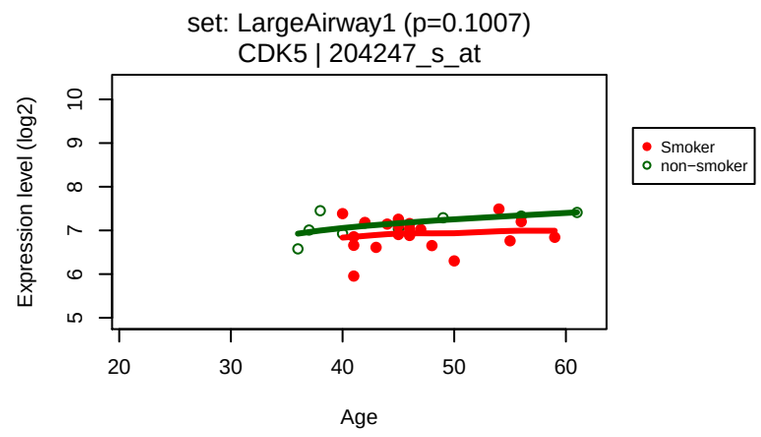

Gene expression of smokers and nonsmokers
gene: CDK5 | 204247_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2756
p=0.7474
p=0.995
p=0.1007
p=0.795
p=0.0262
p=0.5188
p=0.006
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

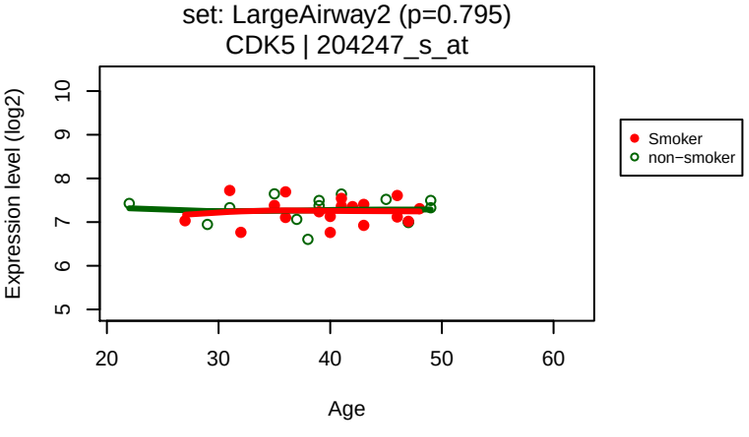

Gene expression ratio (smokers/nonsmokers)
gene: CDK5 | 204247_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2756
p=0.7474
p=0.995
p=0.1007
p=0.795
p=0.0262
p=0.5188
p=0.006
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

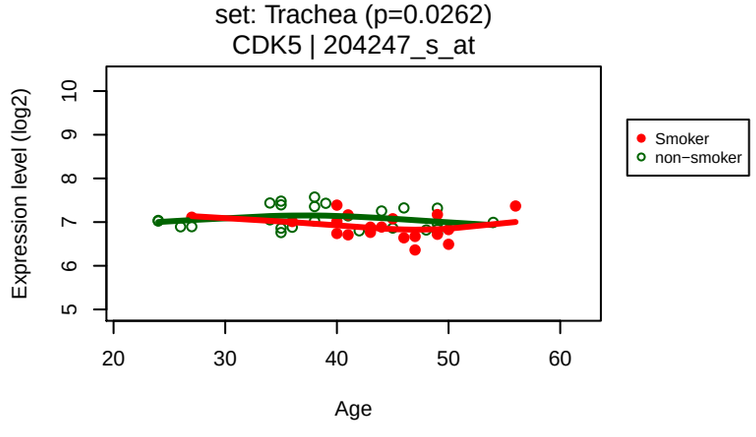

Gene expression ratio in Males
gene: CDK5 | 204247_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8942
p=0.6752
p=0.043
p=0.0105
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CDK5 | 204247_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5698
p=0.3216
p=0.0258
p=0.2728
SmallAirway2
LargeAirway2
Trachea
Macrophage2

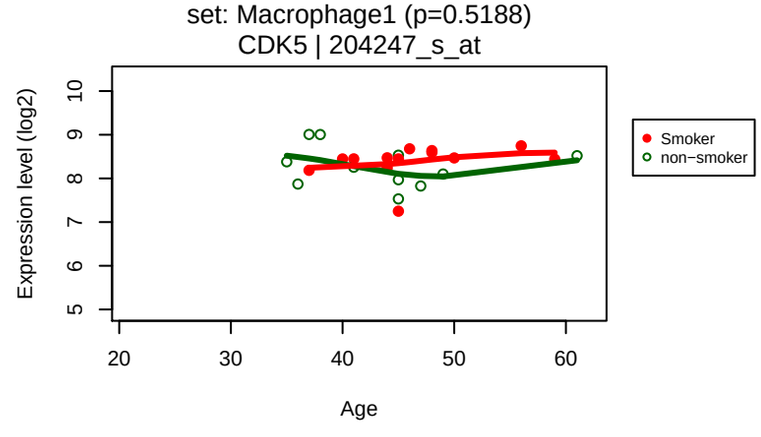

set: SmallAirway1 (p=0.2756)
CDK5 | 204247_s_at
Expression level (log2)
Age
Smoker
non−smoker

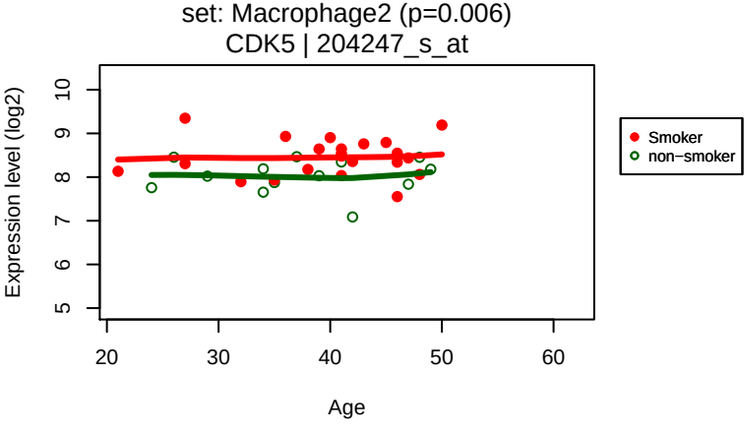

set: SmallAirway2 (p=0.7474)
CDK5 | 204247_s_at
Expression level (log2)
Age
Smoker
non−smoker

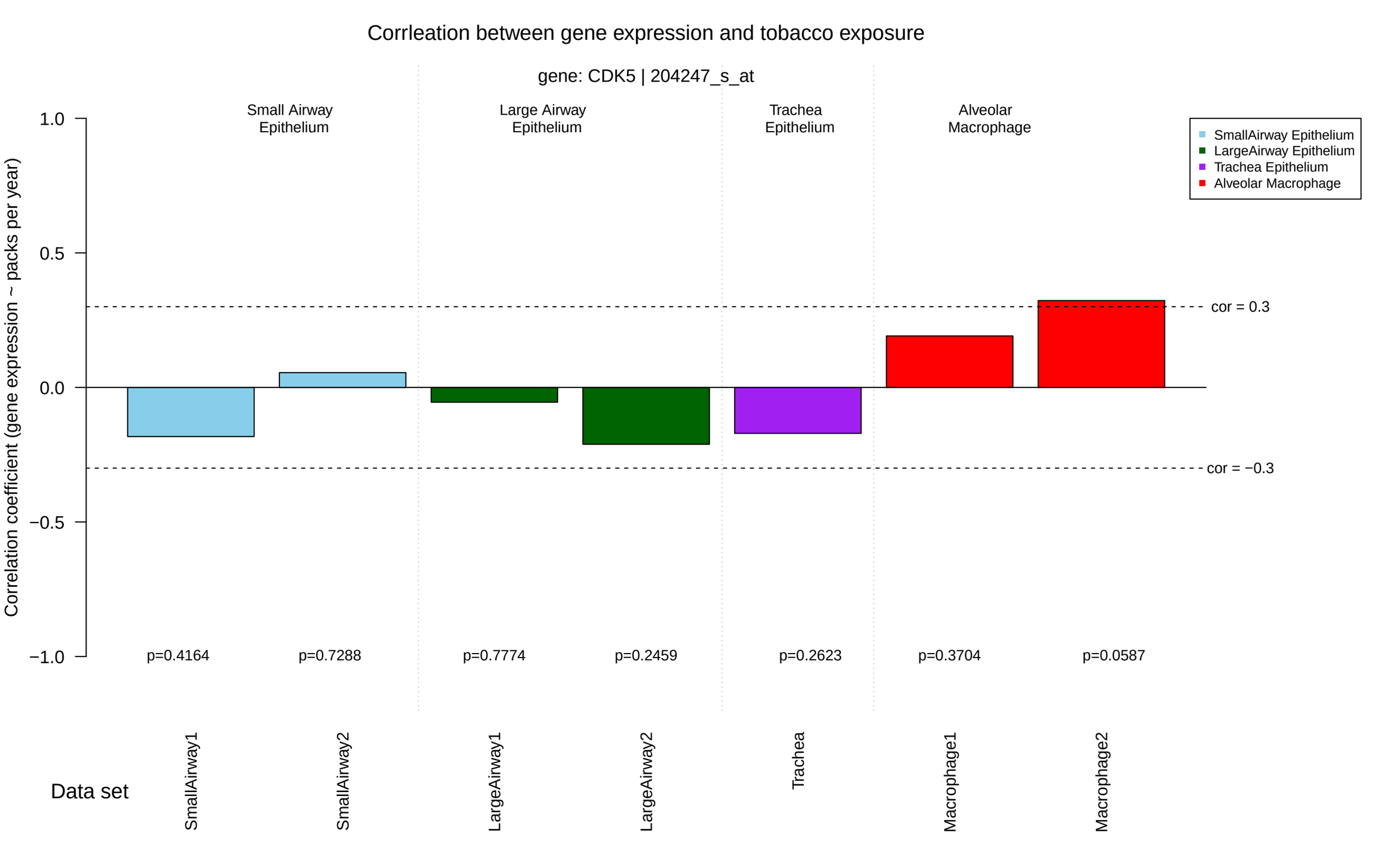

set: LargeAirway1 (p=0.1007)
CDK5 | 204247_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.795)
CDK5 | 204247_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0262)
CDK5 | 204247_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5188)
CDK5 | 204247_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.006)
CDK5 | 204247_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CDK5 | 204247_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4164
p=0.7288
p=0.7774
p=0.2459
p=0.2623
p=0.3704
p=0.0587
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2